

# Technical talk

Math, Diagrams, and Executable Code

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# Roadmap

1. Formalize the decision problem
2. Expose the computation
3. Explain system structure
4. Connect results to evidence

# Model

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# Sequential decisions under uncertainty

We consider an agent that repeatedly observes a state, chooses an action, and receives a reward.

- **State:**  $s \in \mathcal{S}$  summarizes the information available now.
- **Action:**  $a \in \mathcal{A}$  changes both reward and the next-state distribution.
- **Objective:** maximize expected discounted return over an indefinite horizon.

## Modeling assumption

The current state is sufficient: conditional on  $s_t$  and  $a_t$ , the future is independent of the earlier history.

# Bellman optimality equation

$$V^{\star(s)} = \max_a R(s, a) + \gamma \sum_{s'} P(s' \mid s, a) V^{\star(s')}$$

The recursion separates immediate utility from the expected value of all subsequent decisions [1].

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# Computation

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# An executable analysis

The source below is executed during compilation. Its figure is cached by Calepin and placed on the following slide.

```
library(ggplot2)
fit <- lm(mpg ~ hp + factor(am), data = mtcars)

ggplot(mtcars, aes(hp, mpg, color = factor(am))) +
  geom_point(size = 2.6, alpha = 0.85) +
  geom_smooth(method = "lm", se = FALSE, linewidth = 0.9) +
  scale_color_manual(
    values = c("0" = "#657377", "1" = "#f28e2b"),
    labels = c("Automatic", "Manual")) +
  labs(x = "Horsepower", y = "Fuel efficiency (mpg)", color = "Transmission") +
  theme_minimal(base_family = "Fira Sans", base_size = 13) +
  theme(legend.position = "top", panel.grid.minor = element_blank())
```

# Horsepower predicts lower efficiency

- Fuel efficiency declines as horsepower increases.
- Transmission groups occupy different regions of the sample.
- The fitted lines summarize association, not causation.

*Data: mtcars, 32 model-year 1973 to 1974 automobiles.*

# Computed model summary

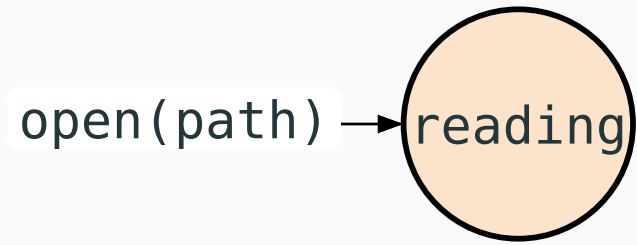
The same R session retains `fit`, so numerical claims can be generated rather than copied into the deck.

```
b <- coef(summary(fit))
r2 <- summary(fit)$r.squared
cat(sprintf(
  "#strong[Estimated association] \
  Holding transmission fixed, an additional 10 horsepower is associated
  with %.2f fewer mpg*. The model explains %.1f%%* of the observed
  variation in fuel efficiency.",
  -10 * b["hp", "Estimate"], 100 * r2
))
```

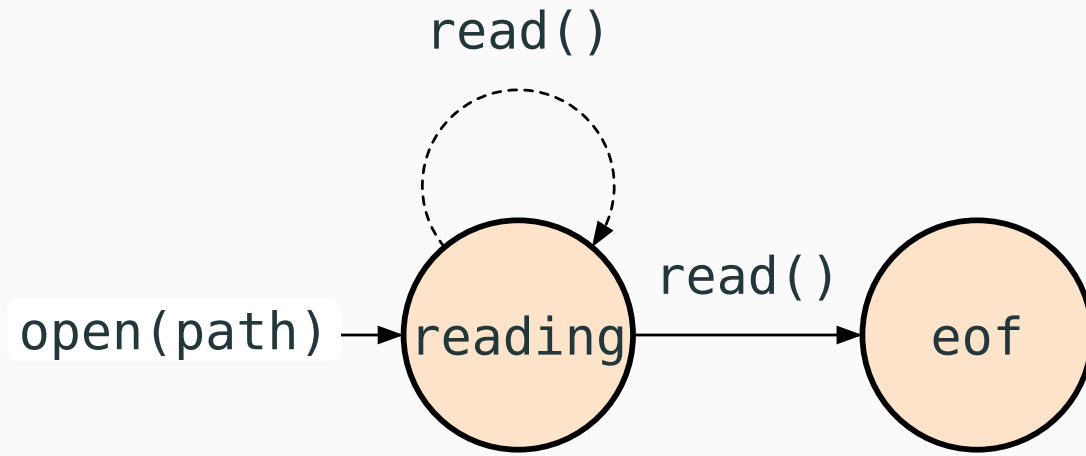
# Structure

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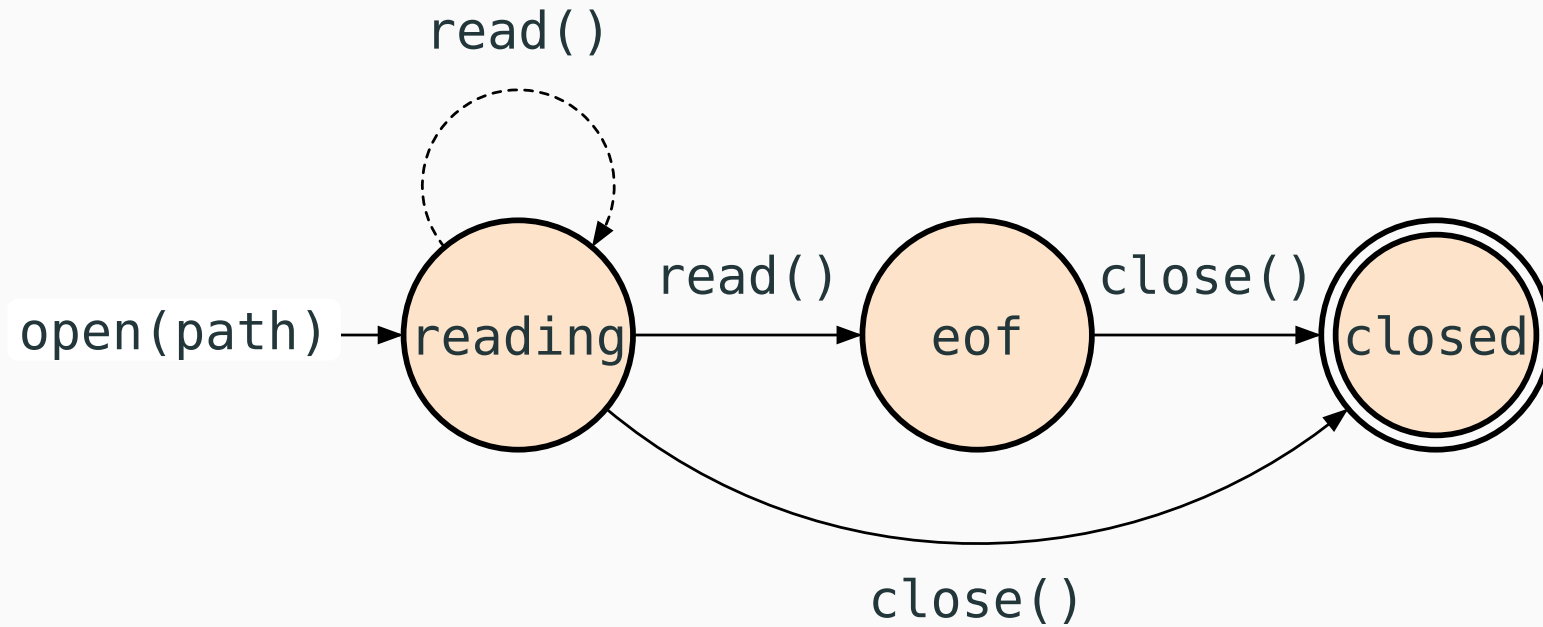
# File-reader state machine



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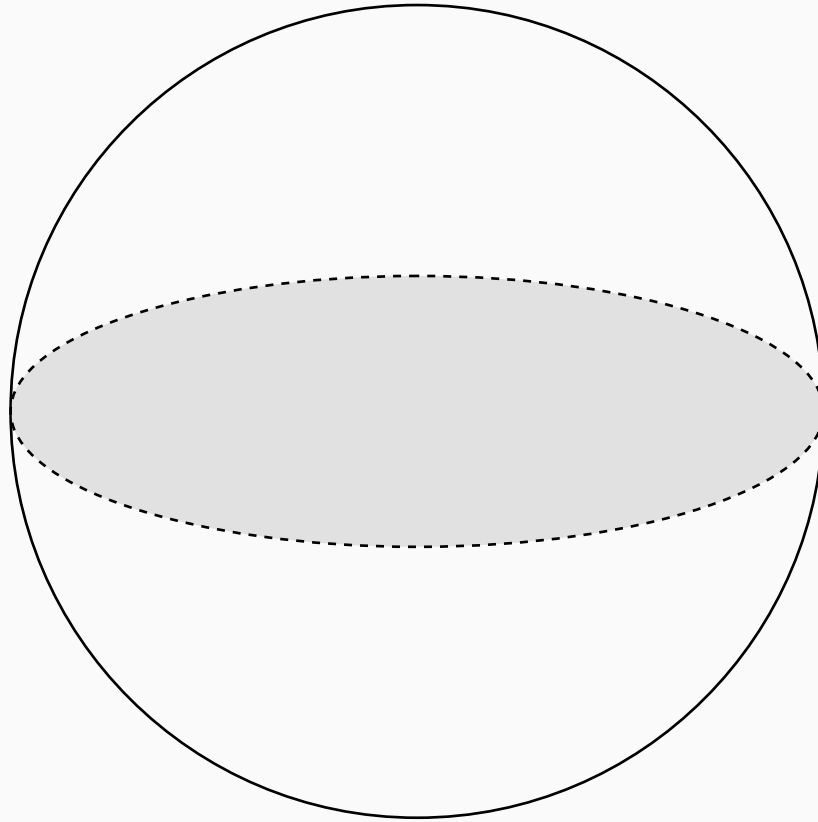
# File-reader state machine



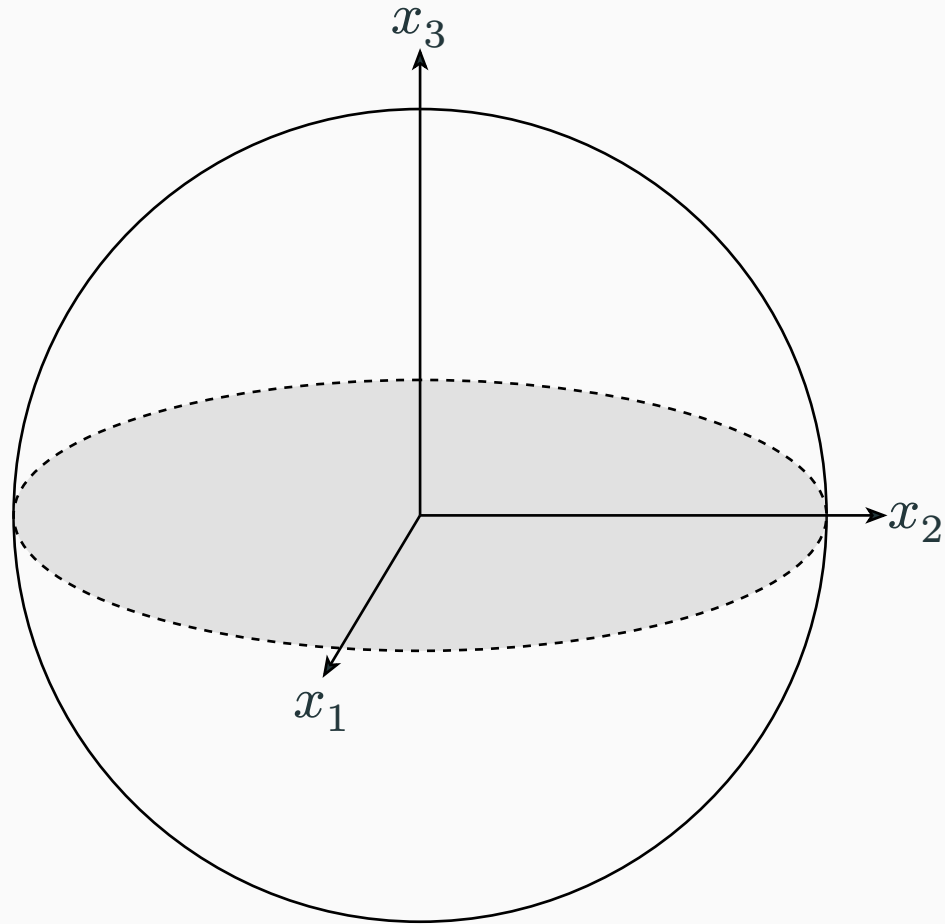
# Geometry

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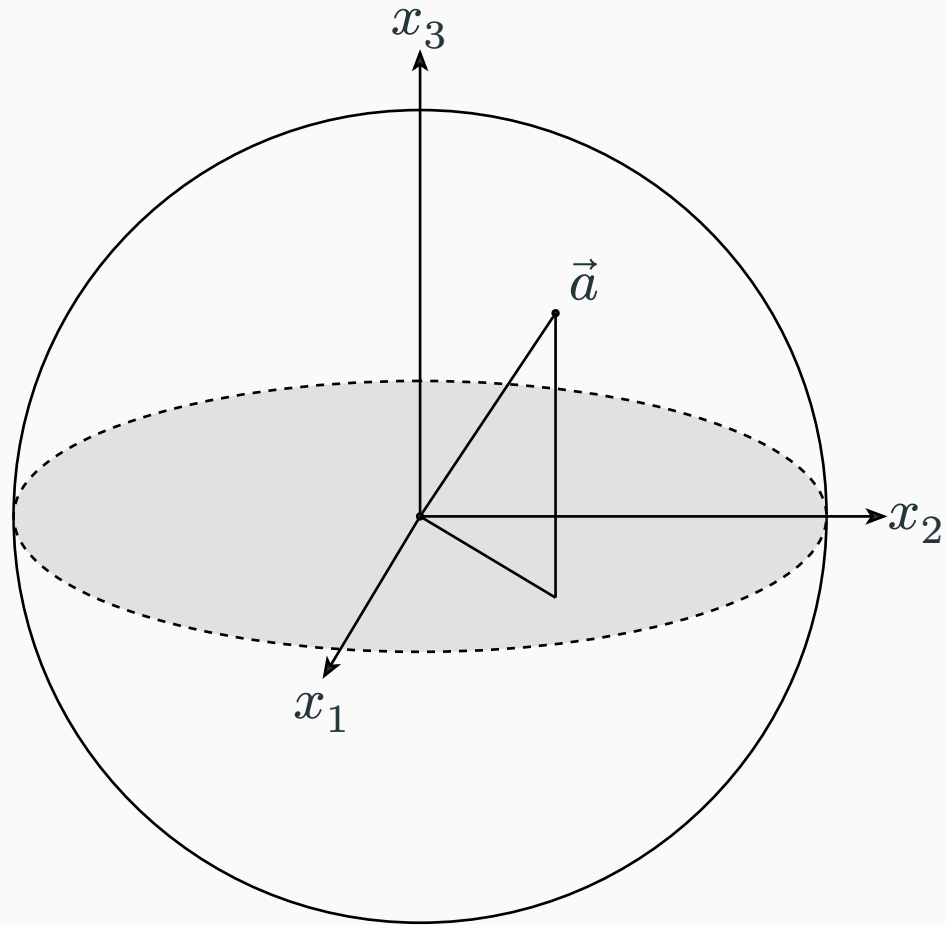
# A qubit as a Bloch vector



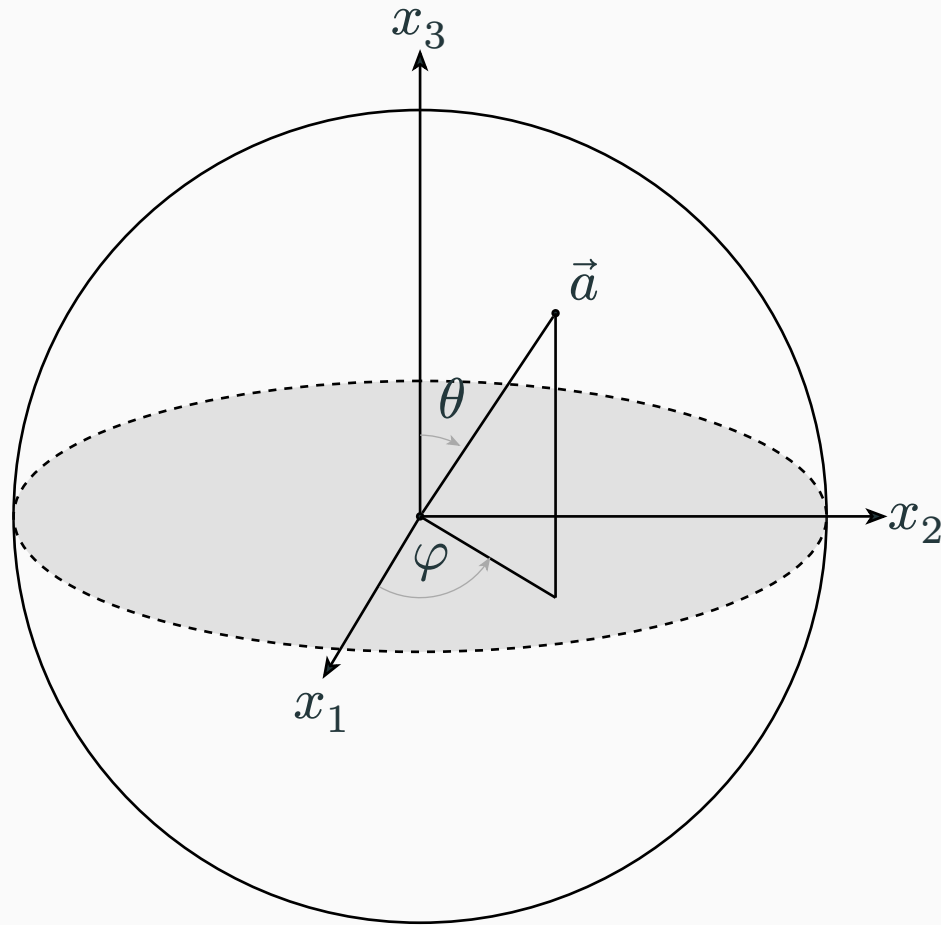
# A qubit as a Bloch vector



# A qubit as a Bloch vector



# A qubit as a Bloch vector



# Evidence

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# What makes a technical slide?

## Include

- a question or claim
- units and definitions
- reproducible computation
- a stated interpretation

## Avoid

- unlabelled decorative plots
- numbers copied by hand
- animation without exposition
- citations disconnected from claims

Questions?

# References

- [1] R. Bellman, *Dynamic Programming*. Princeton University Press, 1957.